🔹 Project Title

Proactive Carbon Intelligence:
ML-Based Emission Monitoring & Anomaly Detection

🎯 Core Idea (MOST IMPORTANT)

Environmental Data
↓
Predict Power Output (ML Model)
↓
Estimate CO_2 Emission
↓
Compare Actual vs Expected
↓
Detect Anomalies

👉 Say this if confused — this is your backbone.

🔹 Why This Project?

- Industrial emissions are high
- Existing systems are reactive
- Need predictive + intelligent monitoring

👉 Keyword: **"Proactive monitoring"**

🔹 Dataset

- Combined Cycle Power Plant dataset
- Features:
    - AT (Temperature)
    - V (Vacuum)
    - AP (Pressure)
    - RH (Humidity)
    - PE (Power Output)

🔹 Key Formula

CO_2 = PE × 0.82

👉 Used because dataset doesn't contain emission values.

🔹 Models Used

1️⃣ Linear Regression

- Simple baseline
- Assumes linear relationship

2️⃣ Gradient Boosting ✅ (FINAL MODEL)

- Captures non-linear relationships
- Lower error → better performance

🔹 Results (REMEMBER THESE)

Linear Regression RMSE ≈ 4.47
Gradient Boosting RMSE ≈ 3.80 ✅ better

👉 Always say: **"Gradient Boosting performed better"**

🔹 Anomaly Detection

Method:

Residual = Actual CO_2 − Predicted CO_2

Model:

Isolation Forest

Result:

192 anomalies (~2%)

🔹 What is Residual?

Difference between actual and predicted value

👉 Used to detect abnormal behavior.

🔹 What is RMSE?

Measures average prediction error
Penalizes large errors more

👉 Simple line:

> RMSE tells how far predictions are from actual values.

🔹 Why Gradient Boosting?

Handles non-linear relationships
Lower RMSE → better accuracy

🔹 Why Not LSTM?

Dataset has no real time dependency
Timestamps were synthetic

👉 This answer = very impressive

🔹 Anomaly Detection Meaning

Detects unusual emission behavior

Examples:

- emission spike

- inefficient operation
- abnormal system behavior

🔹 What You Have Completed

✔ Data preprocessing
✔ CO_2 estimation
✔ Prediction models
✔ Model comparison
✔ Anomaly detection
✔ Visualization

🔹 What Is Pending

❌ Optimization module
❌ Streamlit dashboard

🔹 Future Work

- Optimization of emissions
- Interactive dashboard
- Real-time data integration

🔹 One-Line Summary (VERY IMPORTANT)

👉 If stuck, say this:

> We developed a machine learning system that predicts power output, estimates CO_2 emissions, and detects abnormal emission patterns using anomaly detection.